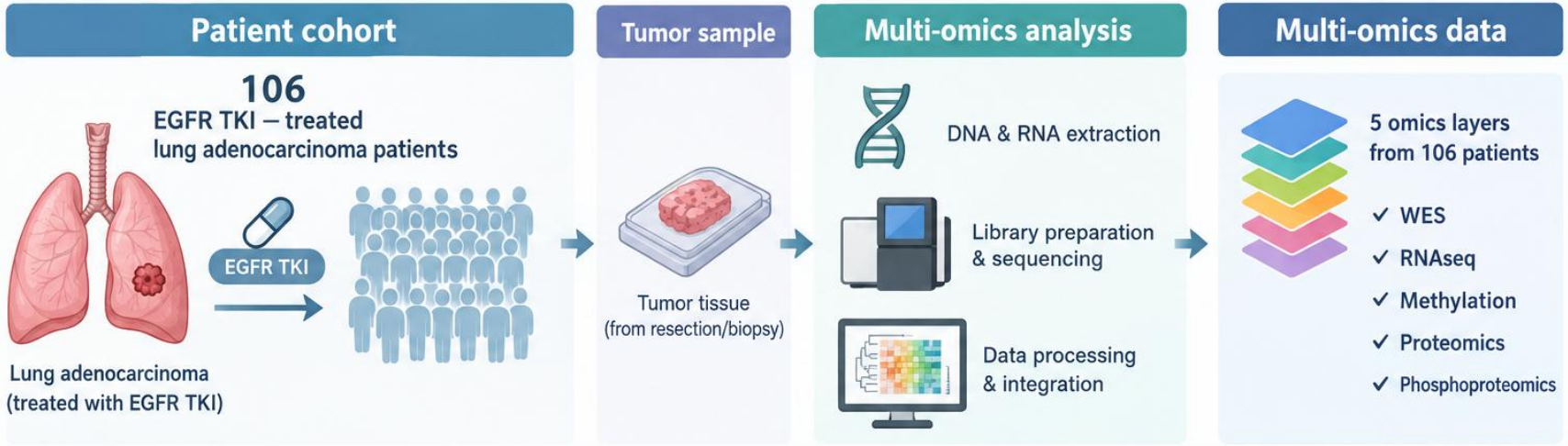
Final Presentation
biovibe-oncology_slyu's-lab

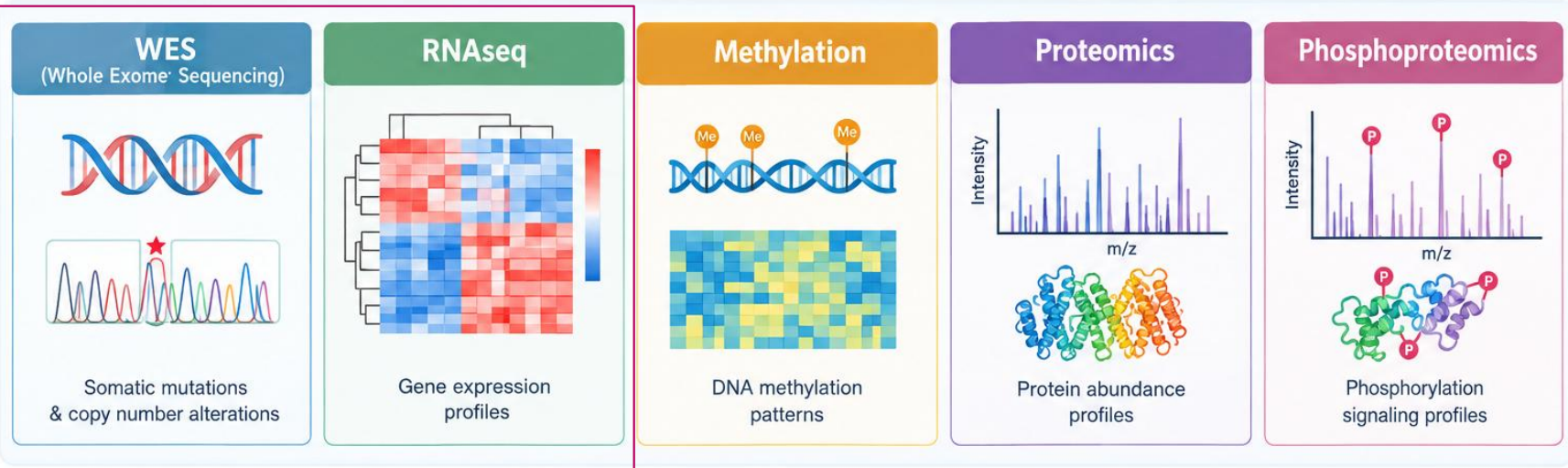
A Biologist-Centric AI Workflow for Decoding EGFR-TKI drug Resistance and Discovering New Drugs.

OpenACC
More Science, Less Programming

11/04/2026

Introduction : EGFR TKI Multiomics





126 mutated protein found in poor outcome patients

X 336 proteins (EGFR TKI resistant library)

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Investigating *De Novo* PPIs and Discovering Novel Drug Targets

Conventional Team Structure (5-7 members)

To build and deploy the pipeline from scratch in **6 months,** a cross-functional team is required:

Bioinformatician : Manages Parabricks, mutation libraries, ESM/AlphaMissense data pipelines, and screening logic.

Computational Chemist/Structural Biologist : Tunes Boltz-2/Chai-1, resolves protein structures, and leads DiffDock and MM-GBSA calculations.

AI/ML Engineer : Deploys generative models (GenMol, MegaMolBART) for compound generation and ADMET filtering.

Cloud Engineer : Configures HPC environments, optimizes H200 multi-node scheduling, and maintains data flow.

Project Manager : Drives timeline and coordinates seamless data handovers across the team.

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Can A biologist with ZERO coding skills build this pipeline?

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Can A biologist with ZERO coding skills build this pipeline?



Yes.
AI agent & Vibe Coding makes it possible.

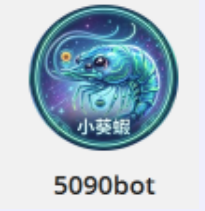
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- **Workflow Speedup by AI Agent_5090bot**

Speed up



OpenClaw: Minimax/Gemini



Pipeline & MVP: 11 days

5x180 / 1x11 = ~81X

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Project Manager : Drives the 3-month timeline and coordinates seamless data handovers across the team.

De novo PPI & molecular drug discovery

Mutation / de novo PPI

318
Parabricks
30-45 min/sample/RTX5090

2,774
AlphaMissense
3 min

889
ESM-2
~40min/H200

596 PDBs
ESMfold
98.2 min/H200

1,685 pair
(5 mut MVP test)
Boltz-2
3.5 hrs / H200

Candidate 126 mut
& 336 TKI signaling library
Chai-1
23 hrs / H200

7 pair
Chai-1_wt 篩選

1
De novo PPI
Mut A × B protein

De novo molecular drug

1,000
GenMol
20-30secs/H200

857
ADMET/RDKitFilter
3 min

Top 10 molecular
DiffDock
~1.4 sec/molecule

600 molecular
MegaMolBART
~1 min/H200

Top 1 molecular
Chai-1 Trimer

1 Molecular x
1 Ligand x 1 Receptor
MM-GBSA
12-13 hr / H200

Multi-Agent Collaboration

Design Principles: Task Allocation · Heterogeneous Computing Resource Scheduling · Traceable Execution Logs · Cross-Agent Collaboration

Computers | Human-in-the-Loop



Task Understanding · Instruction Parsing · Task Dispatch



SSH | Tailscale

Security · Encryption · Cross-Network Connectivity

RTX5090 Laptop



NCHC GPU VM

H200 x1



OpenClaw: Kimi-K3/Claude

H200 x2



Code Execution · Data Analysis · Model Inference

HPC | High-Performance Computing



Notion Collaboration Workflow | Shared Task Management

Legend

- Remote / Control
- Task Execution
- Large-Task
- Shared Workflow

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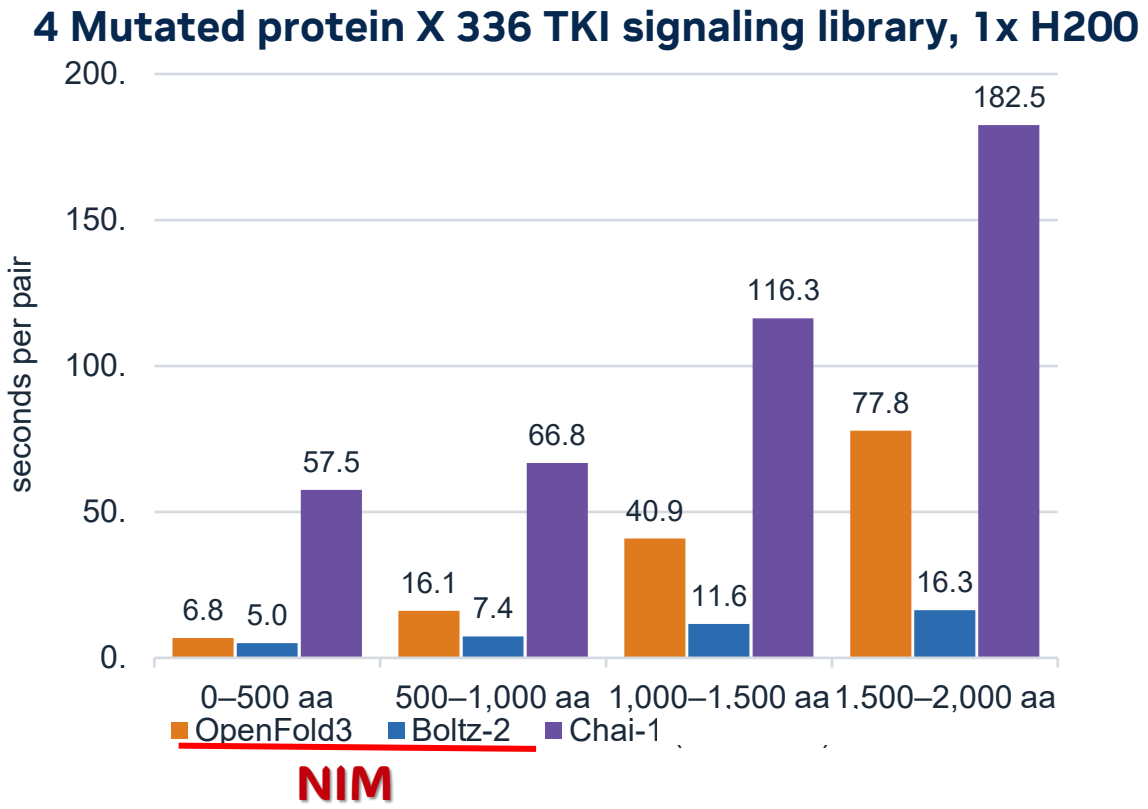
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MegaMolBART
~1 min/H200

Top 1 molecular
Chai-1 Trimer

1 Molecular x
1 Ligand x 1 Receptor
MM-GBSA
12-13 hr / H200

Profiler Output: Speed

Speed up Per pair: **Boltz-2 ≈ 9 s (~10X)**, OpenFold3 ≈ 26 s, Chai-1 ≈ 84 s



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Comparison within OpenFold3, Boltz-2 and Chai-1

iPTM Rank	OpenFold3 NIM	Boltz-2 NIM	Chai-1 50 timesteps*	Chai-1 200 timesteps*
UXXXX_D111N × CDKN2B	**0.051** **1,319**	**0.403** **178**	**0.622** **2**	**0.651** **1**
UXXXX_D111N × CIDEB	**0.106** **768**	**0.648** **12**	**0.563** **4**	**0.473** **31**
UXXXX_D111N × CDKN2C	**0.367** **3**	**0.098** **1,255**	**0.293** **41**	**0.340** **84**
UXXXX_D111N × CARD9	**0.089** **1,021**	**0.149** **1,093**	**0.144** **143**	**0.265** **164**

*num_diffn_timesteps

iPTM:

> 0.7 High Confidence: High-reliability interface
0.5 – 0.7 Moderate Confidence: Potential weak binders
< 0.5 Low Confidence: Unstable interface prediction

Ref. FoldBench（GitHub，2025-12 update）

Success Rate	Protein-Protein	Protein-ligand
Chai-1	66.95%	49.28%
OpenFold 3 (preview)	68.22%	40.85%
Boltz-2*	**70.54%**	**53.90%**

For protein–ligand interfaces: LRMSD < 2 Å and LDDT-PLI > 0.8
For all other interfaces: DockQ ≥ 0.23

Benchmarking all-atom biomolecular structure prediction with FoldBench

Nature Communications volume 17, Article number: 442 (2026)

Nsight analysis | Three bottleneck in Chai-1

① Data Transfer

0.92 GB/s

GPU Host-to-Device (DtoH) transfers occurred 218,295 times, totaling 6,289 GB over 6,841 seconds (~22 sec/ppi). The figure shows the transfer-back speed; Host-to-Device (HtoD) transfer rate is ~5.7 GB/s with a total transfer volume of 6,383 GB, nearly identical to the transfer-back volume.

② Computing Precision

14.7 %

BF16 computations on Tensor Cores account for only 14.7% of kernel time; FP32 matrix multiplication accounts for 27.5%, and LayerNorm/Softmax account for 29.3%.

③ GPU Idle

1,585 segments

Gaps exceeding 1 second total 1,585 segments (~5 sec/ppi), accounting for 97% of non-kernel time.

Profiler Output: Speed

- **Workloads Speedup by GPU**

31,672 PPI	Chia-1	Chia-1 batching	Boltz-2 NIM	Boltz-2 BioIR
1x H200	~2,024 hr	~675 hr	197 hr	145 hr
8x H200 (HPC)	~253 hr	~85 hr	27 hr	18 hr



Lessons Learned

1. **Models & NIM:** Robust validation & microservice acceleration.
2. **GPU Optimization:** Nsight profiling + dynamic batching for peak efficiency.
3. **HPC Scalability:** Large-scale compute infrastructure for heavy workloads.
4. **BioNeMo Inference Runtime (BioIR) :** Maximizing GPU utilization on H200 clusters.

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BioVibe-Onco: A Biologist-Centric AI Workflow for Decoding EGFR-TKI drug Resistance and Discovering New Drugs.

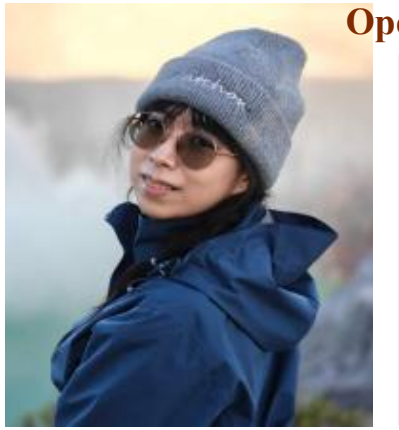


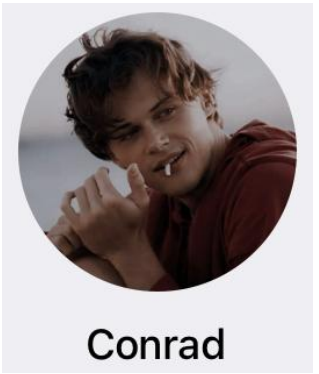
Team Leader (Yi-Jing Hsiao PhD, NTU/Sinica)



Lab PI (Prof. Sung-Liang Yu, NTU)

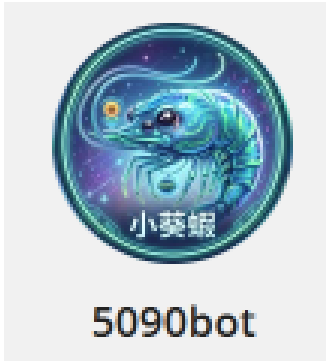
OpenClaw: Kimi-K3/Claude





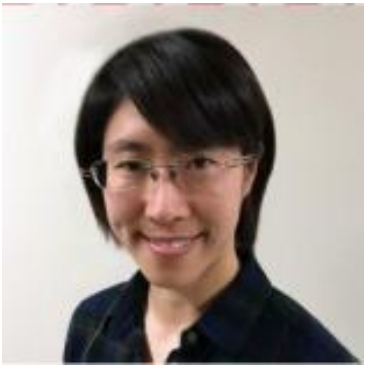
Team Members (FAN-NI HSING, NTU)

OpenClaw: Minimax/Gemini





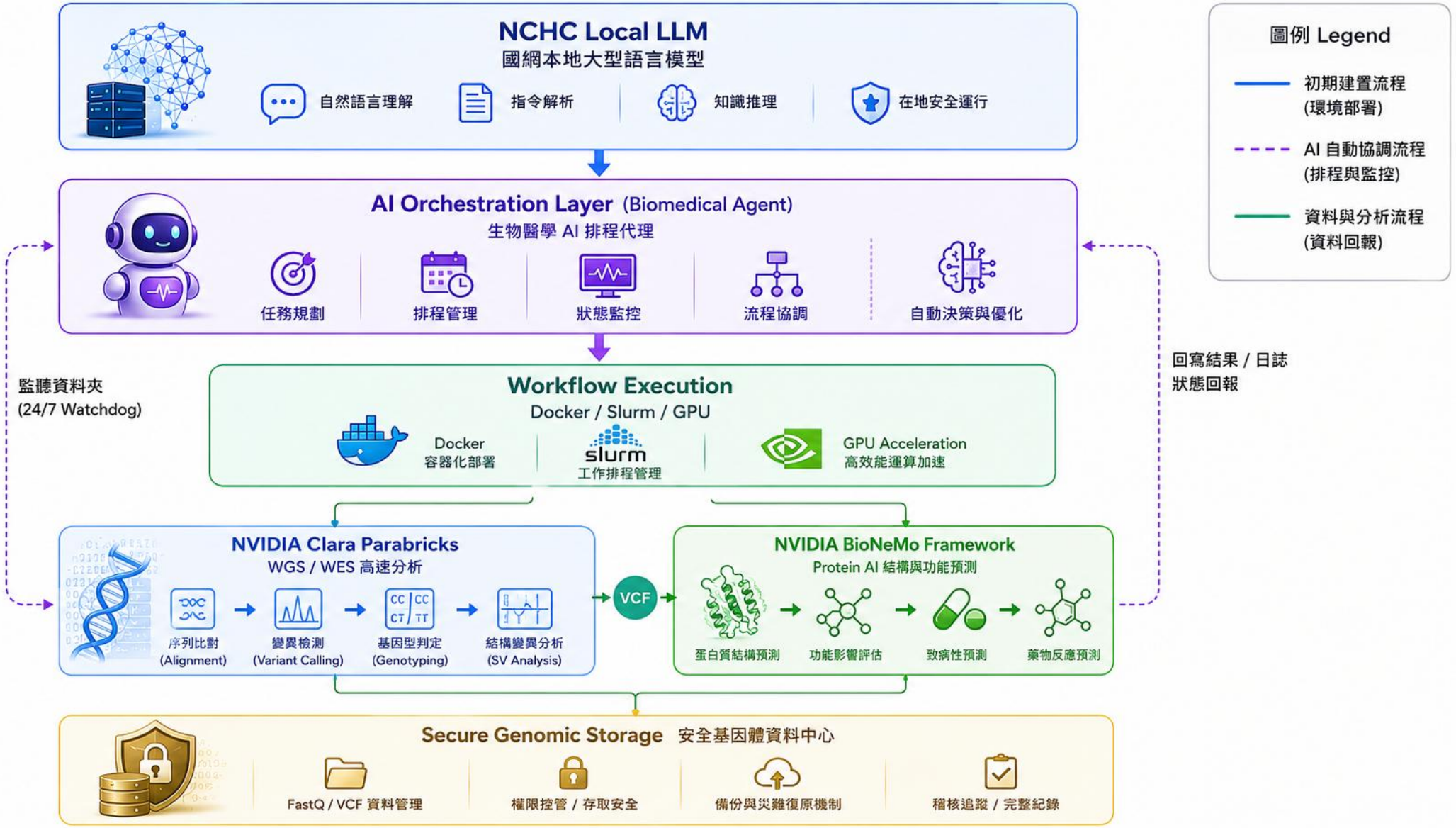
Team Members (小葵蝦/蝦大王/Conrad)



Team Mentors (Ying-Jia Chen PhD, NVIDIA)

Thank you for your attention!!

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NSIGHT 分析 | 三個瓶頸

時間花在資料搬移、運算精度與 GPU 閒置三處

① 資料搬移

0.92 GB/s

GPU 傳回主機（DtoH）共 218,295 次、6,289 GB，耗時 6,841 秒（約每組 22 秒）；上圖為傳回的速度，傳入（HtoD）約 5.7 GB/s；傳入量 6,383 GB，與傳回量幾乎一樣。

推論：搬移時 CPU 在等（API 8,211 秒 ≈ 搬移 7,983 秒），符合未鎖定記憶體；搬移來源為 low_memory 與 ESM（依原始碼）

② 運算精度

14.7 %

Tensor Core 的 BF16 運算只佔 kernel 時間 14.7%；FP32 矩陣乘法佔 27.5%，LayerNorm 與 Softmax 佔 29.3%。

推測，尚未驗證：FP32 矩陣乘法沒有使用 TF32

③ GPU 閒置

1,585 段

超過 1 秒的空檔共 1,585 段（約每組 5 段），佔無 kernel 時間的 97%。

每組固定成本約 53 秒，與閒置 47 秒相符；原因為每組重新載入 ESM 與 6 個元件（依原始碼）

「無 kernel 時間」含記憶體搬移；橘色底的說明由追蹤檔、耗時表與 chai-lab 0.6.1 原始碼推得，需用 A/B 實驗確認